

Part 1: Datasheets for Datasets

1. What exactly is a datasheet for a dataset?

A datasheet is a document that provides detailed information about a dataset, such as why it was created, how the data was collected, what the dataset contains, and how it should or should not be used.

1. Why do the authors think datasheets are important?

The authors argue that datasheets are significant as they make datasets more transparent and easier to understand, especially for those who were not involved in creating them. This information can help researchers interested in the datasets identify limitations and potential risks before using a dataset, making it easier to determine whether a dataset is appropriate for a particular purpose.

2. In what ways was the development process for the guidelines set out by the authors iterative and/or incremental?

The development of the guidelines was iterative because the authors did not create the final set of questions all at once. They developed various questions, “tested” them on datasets, and revised the guidelines. The process was also incremental because the guidelines were built and expanded over time, with new questions and considerations added as the authors identified additional parameters.

3. Pick two of the questions that the authors suggest answering in a datasheet. Why do you think they are asking these questions -- what harms might they be trying to prevent, and what caveats might they uncover?

- a. The question, “What processes were used to collect the data?” provides researchers with the methodology involved in the data collection within the dataset. This provides researchers with the ability to ensure their data aligns with the focus of their research. For example, if data were collected from only a geographic area or demographic group, researchers using the data might incorrectly assume that the dataset represents a much broader population than it actually does, thus drawing incorrect conclusions in their results.
- b. The question, “Are there any errors, sources of noise, or redundancies in the dataset?” is significant because flaws within a dataset can affect the results of research built from it. Researchers being provided with information about missing information, duplicate data, or errors allows them to account for those limitations within their projects or research.

Part 3: Debugging Conditionals

1. pick_dinner: This function can raise an UnboundLocalError. Your job is to find the inputs that cause this error, explain why the error happens, and correct the function so that it cannot raise the error.

The input that is causing this error is (False, False). An UnboundLocalError occurs within a function when a variable is read before it has been assigned a value within the function. In this example, this error occurs because dinner is not defined when both inputs are False. So, we’re effectively calling a variable that has not been defined, so Python does not know what to do with the variable.

2. `quadratic_solver`: This function can raise a `ValueError` when given certain inputs. Please characterize the inputs that cause this error, explain why it's raised, and add input validation to the function.

The function raises a `ValueError` when given a set of inputs (a,b,c) that yields a negative output. This error is raised because the square root function domain contains strictly nonnegative numbers; thus, it cannot compute negative discriminants.

3. `weather_gear`: This function does not behave according to its docstring in some significant ways. Please explain at least two ways that it does not reflect what the docstring says it should be doing, then correct the function. HINT: It's okay if the function is not pure once you've modified it.

The if statement on Line 22 of the file `weather_gear.py` is incorrect. This line states that if the temperature is less than 0 and simultaneously greater than 100, to return an invalid error message string. However, this is semantically incorrect, as this condition will never be satisfied. Thus, any input that the function receives for temperature will be passed into the function regardless of its validity.

Another instance where the code does not reflect the docstring in a significant manner is when the initial temperature condition is met, while it's raining and also sunny; the program does not return a string that provides a recommendation for all conditions.

Part 5: Write your own Homework Problem

This problem is designed to assess students' ability to use functions to organize a larger Python program. This seems to be a skill that the class, including me, seems to struggle with. The validation function requires students to use Boolean logic with multiple conditions and return either `True` or `False`. The driver code then has to use that returned value to determine whether the rest of the program should run. The most challenging part for an average student will likely be understanding that `validate_event()` should not print the result itself. Instead, it should return information to the driver code, which then decides what to do with that information. Students must also correctly use Booleans to prevent invalid inputs from reaching the calculation functions. In completing this problem, we should be able to understand to a greater extent how individual functions interact with one another and how validation can control the flow of an entire program.